Explain Network and Computer Attacks, Focusing on Malware

Introduction

A **network and computer attack** is any attempt by an attacker to damage, disrupt, steal data from, or gain unauthorized access to a computer system or network. These attacks can target individual computers, servers, mobile devices, or entire organizational networks.

One of the most common and dangerous forms of such attacks is **malware**. Malware stands for *malicious software*. It is designed to enter a system without permission and perform harmful actions such as stealing data, spying on users, deleting files, or blocking services.

What Are Network and Computer Attacks?

Network and computer attacks aim to:

- Break into systems
- Steal confidential information
- Disrupt services
- Spread harmful programs
- Take control of machines

These attacks may come through:

- Email attachments or links
- Fake websites
- Infected software downloads
- USB devices
- Network vulnerabilities
- Weak passwords

Malware & its types

Malware is any software intentionally created to harm a computer or network. It can:

- Corrupt files
- Spy on users

- Steal passwords
- Encrypt data
- Slow down systems
- Allow attackers remote control

Malware is often hidden inside normal-looking files or programs so that users install it without knowing.

Types of Malwares

Different types of malwares perform different attacks:

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1. Virus

A **virus** attaches itself to a legitimate program or file. When that program runs, the virus activates and spreads to other files.

Effects:

- Corrupts data
- Slows down system
- Crashes programs

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2. Worm

A **worm** spreads automatically through networks without user action. It exploits security holes and copies itself to other machines.

Effects:

- Consumes network bandwidth
- Spreads very fast
- Can install other malware

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3. Trojan Horse

A **Trojan** looks like a useful program but secretly performs harmful actions once installed.

Effects:

- Creates backdoors
- Steals information
- Gives attacker control

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4. Ransomware

Ransomware locks or encrypts files and demands money to restore access.

Effects:

- Makes systems unusable
- Causes financial loss
- Interrupts business operations

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5. Spyware

Spyware secretly monitors user activities and sends information to attackers.

Effects:

- Steals passwords
- Tracks browsing
- Captures keystrokes

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6. Adware

Adware displays unwanted advertisements and may redirect browsers to unsafe sites.

Effects:

- Annoying pop-ups
- Slows system
- May collect user data

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7. Botnet Malware

This malware turns infected computers into **bots** controlled remotely by attackers.

Effects:

- Used for DDoS attacks
- Spam sending
- Data theft

How Malware Attacks Networks and Computers

Malware usually follows these steps:

1. **Entry** – Comes through email links, downloads, or USB drives.
2. **Installation** – Runs secretly in the background.
3. **Propagation** – Spreads to other systems.
4. **Execution** – Performs harmful tasks like stealing data.
5. **Persistence** – Tries to remain hidden for a long time.

Prevention and Protection Against Malware

To defend networks and computers, organizations use:

1. Antivirus and Anti-Malware Software

Detects and removes harmful programs.

2. Firewalls

Block unauthorized network traffic.

3. Regular Updates

Patch software and operating systems.

4. Email Filtering

Stops phishing emails and malicious links.

5. User Awareness

Train employees to avoid suspicious files.

6. Backup Systems

Allow recovery after ransomware attacks.

7. Intrusion Detection Systems

TCP/IP Model

The **TCP/IP model** is a reference model used to describe how data is sent and received over the Internet and computer networks.
TCP/IP stands for **Transmission Control Protocol / Internet Protocol**.

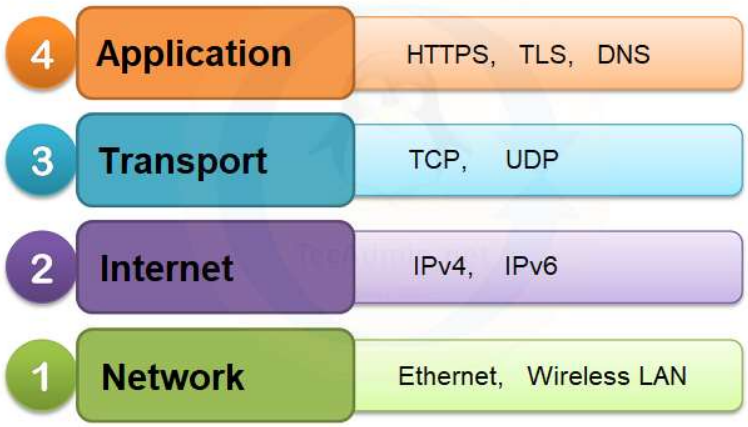
The TCP/IP model explains communication using **four layers**, where each layer performs specific functions.

Layers of the TCP/IP Model

The TCP/IP model has **four layers**:

1. Application Layer
2. Transport Layer
3. Internet Layer
4. Network Access (Link) Layer

Each layer has a clear role in the communication process.



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1. Application Layer

The **Application Layer** is the topmost layer. It provides services directly to users and applications.

- Enables user applications to access the network
- Supports web browsing, email, file transfer, etc.
- Formats and presents data

Example Protocols:

- HTTP / HTTPS – Web browsing
- FTP – File transfer
- SMTP – Sending email
- POP3 / IMAP – Receiving email
- DNS – Name to IP address conversion

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2. Transport Layer

The **Transport Layer** is responsible for end-to-end communication between devices.

Functions:

- Breaks data into segments
- Ensures reliable delivery
- Error checking
- Flow control
- Re-sending lost data

Main Protocols:

TCP (Transmission Control Protocol)

- Reliable and connection-oriented
- Ensures data arrives correctly
- Uses acknowledgements and retransmission

UDP (User Datagram Protocol)

- Faster but unreliable
- No guarantee of delivery
- Used in streaming and online games

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3. Internet Layer

The **Internet Layer** handles logical addressing and routing of packets between networks.

Functions:

- Assigns IP addresses
- Routes packets across networks
- Packet forwarding
- Fragmentation and reassembly

Protocol:

- IP (Internet Protocol)
- ICMP – Error reporting
- ARP – Finds physical address
- IGMP – Multicast communication

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4. Network Access (Link) Layer

This is the lowest layer. It deals with physical transmission of data over the network medium.

Functions:

- Converts packets into frames
- Physical addressing (MAC address)
- Controls access to media
- Error detection at hardware level

Examples:

- Ethernet
- Wi-Fi
- Optical fiber
- LAN technologies

Advantages of TCP/IP Model

- Open standard and widely used
- Scalable for large networks

- Works on different hardware
- Reliable communication
- Supports routing across the globe
- Flexible architecture

Comparison with OSI Model (Brief)

OSI Model	TCP/IP Model
7 Layers	4 Layers
Conceptual model	Practical implementation
Presentation & Session separate	Included in Application layer
Used for teaching	Used in real networks